The Observer Pets
Interview

Life in the old dog yet: how biotech firms are looking to extend the lives of our pets

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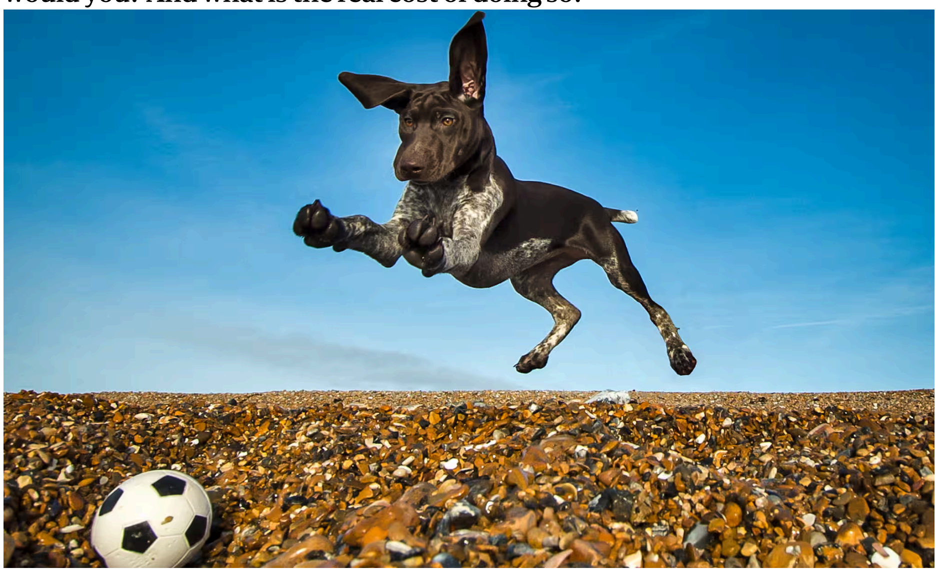
If you could increase the lifespan of your pet dog or cat, would you? And what is the real cost of doing so?



Leap of faith: one biotech startup has already secured $150m in investment to develop a lifespan-extension drug for dogs. Photograph: Brighton Dog Photography/Getty Images

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Last November, my family brought home a puppy. Frankie was eight weeks old when he came to live with us, and right now, watching him bound around with my seven-year-old son, I don't want to imagine ever saying goodbye to him. Well, maybe I won't need to, or rather, I can at least kick that day into the long grass, and buy Frankie some extra time. After all, scientific understanding of the mechanisms of ageing has never been better; there is a plethora of longevity products to choose from and more in the pipeline, including a kind of diet pill for dogs; and, thanks to research into lifespan expansion for pets over the last decade, That's fine by me – let the cycle of life and death proceed apace. I side with the dog owners who are happy to take however many woofs, walks and belly rubs they can get. But the figures show that many of us are investing a significant amount of money and time into extending the lifespan of our four-legged friends. In the UK, the pet supplement industry is worth around $255m, with the overall pet market in the US set to reach a value of $200bn by 2030: this growth is driven partly by an increased spending on pharmaceuticals, diagnostic testing and vet bills. There are wearable activity trackers for dogs and cats, smart collars that collect sleep data, temperature-adjustable beds and apps to analyse your pets' poop. But we're now seeing the emergence of something new: longevity-focused pills and injections that can address the more fundamental causes of pet mortality.

"We live in the same environments as our dogs and share similar lifestyles," says Celine Halioua, the founder of Loyal – a San Francisco-based biotech startup that's so far secured $150m in investment to develop what it hopes will be the first widely available lifespan-extension drug for dogs.

Loyal's research is based on the observation that larger dog breeds tend to have shorter lifespans – Newfoundlands live for an average of 8-10 years, while Chihuahuas average 15-17. To address this, they're developing a range of vet-prescribed products for the US market that turn down production of IGF-1, a hormone that manages cell growth, found at high levels in larger dogs. Research across a number of species suggests links between IGF-1, age-related disease and longevity in several species – driven in part by higher oxidative stress and chronic inflammation.

> ***Intervention's only worthwhile if it increases an animal's healthspan; the proportion of life they're active and healthy***

"You could argue that by breeding for larger dogs, we have – to some extent – also bred in a faster rate of ageing in those dogs," says Dr Matt Kaeberlin, a biologist and longevity expert. "And so turning down IGF-1 would almost definitely have an effect if you administered it while the dog was young – but then, of course, you'd end up with a small dog. The question is if it will work in middle-age, after the dog has stopped growing. But even if it doesn't, it should have a pretty big effect on cancer [IGF-1 stimulates cell proliferation, which is linked to the development of some cancers], which we know is a major killer." As dogs live longer, cancer is increasingly common as a cause of death – especially among breeds like the Bernese mountain dog and bull mastiff. "That means Loyal really has two chances for their system to work," says Kaeberlin. "So I'm pretty optimistic that it's going to show some positive effects." The company has three drugs in development – Loy-001 and 003 are specifically aimed at larger dogs, while Loy-002, aimed at dogs over 10, is the closest to full FDA approval. In February, the daily tablet received its "reasonable expectation of effectiveness" acknowledgment from the FDA and could be prescribed in the US this year.

Of course, this isn't the only theory about how pets' lives might be extended, or the only solution on offer. One of the best-evidenced interventions for humans and other species is calorie restriction: in one 14-year study conducted in a laboratory environment, Labrador retrievers that were fed 25% less than their full-fed siblings saw a mean increase in lifespan of 15% (important to note: the "lean-fed" dogs weren't malnourished and the diet restriction had no adverse effects on skeletal strength or metabolism). Similar levels of calorie restriction seem to have a positive effect on biomarkers of ageing in humans – but very few people are suggesting that it's something you should try with a pet.

"Caloric restriction in a laboratory environment is very protected," says Dr Kaeberlin. "Animals' exposure to pathogens in a lab is very different and the number of processes in the body that are modified by restricting calories is in the tens of thousands. But there are a lot of negative consequences that go along with very severe caloric restriction in the real world." In other words, it's fine, and probably advisable, to keep an eye on your dog's weight – obesity in animals, as in humans, can lead to a raft of health conditions – but not to deliberately restrict their eating over the long term. However, a project co-created by Kaeberlin is also testing another intervention that mimics some of the effects of calorie restriction and can actually be administered in pill form – an artificial drug named rapamycin, originally developed as an immunosuppressant for organ transplant patients.

The Dog Aging Project, a US-based research initiative currently following tens of thousands of non-laboratory dogs in a long-term study, has been running trials to determine the effects of rapamycin on dogs since 2018, after receiving a five-year grant for nearly $29m from the National Institute on Aging. They've prescribed rapamycin to thousands of pet dogs in double-blind, placebo- controlled trials – tracking their health progress through annual questionnaires (or in some cases, DNA testing).

"I believe rapamycin is the best shot on goal," says Kaeberlin. "Other than caloric restriction, it's the intervention for which there is by far the most data in preclinical studies, both for lifespan and pretty much every measure of healthspan that people have looked at." Rapamycin has already shown promising results in worms, yeast and mice – in the latter, it's been shown to increase lifespan by a median 25%, while also affecting the prevalence of certain cancers.

> ***Whatever genetic hand a pet's been dealt, stress, nutrition and exercise all play a part***

Some of Silicon Valley's human longevity fans, not wanting to wait for formal human trials or FDA approval, are already including rapamycin in their supplement stacks. (Bryan "Don't Die" Johnson, the Silicon Valley entrepreneur who spends $2m a year on longevity treatments and has his own Netflix documentary, recently ended his own five-year protocol, citing side effects including "soft tissue infections, lipid abnormalities… and increased resting heart rate".) Rapamycin works by inhibiting a protein now commonly known as mTOR (or the "mechanistic target of rapamycin" – yes, it's named after the drug), which regulates cell growth and metabolism. "What mTOR fundamentally does is that it senses the environment an animal is in, particularly with regard to nutrition, and helps the animal's cells make a decision about whether it's a good time to grow or reproduce," says Kaeberlin. "So for instance, if there's not very much food around that's a really bad time to have babies – so that turns mTOR down. And one of the side effects of that from an evolutionary perspective is that you enhance stress resistance – and one consequence of that increased resistance is slower biological ageing."

Other competing theories might offer promising leads, but aren't (yet) as well funded or researched. The New York-based Vaika Project has spent years tracking the health of a group of 103 retired sled dogs, looking at DNA damage, but doesn't yet have a suggested intervention in pill form and has been forced to cut back its tracking efforts due to funding issues. In Japan, former professor and immunology researcher Toru Miyazaki is studying an inhibitor that might protect cats from kidney disease by helping them dispose of dead cells more efficiently – AIM30, a cat kibble based on his findings, is already on sale, but trials for a vaccine are ongoing. At least one breed of guinea pigs has had their entire genome sequenced as a model for studying Alzheimer's, heart disease and other disorders in humans – but so far, that hasn't resulted in any longevity interventions for the pet variety. So what about owners who want to tackle their beloved pets' health now?

Plenty of over-the-counter supplements offer ingredients that have shown promising results in mice or flies, but little evidence of their efficacy in larger animals. Humans are more helpful: there are any number of other lifespan or healthspan-enhancing interventions available to certain pets, some better evidenced than others. DNA testing, for instance, can help to predict whether your cat or dog has genetic mutations that might predispose them to certain illnesses or adverse reactions to other medication. Epigenetics is another promising area for investigation, as studies increasingly suggest that environmental factors can play a key role in controlling which of an animals' existing genes are expressed - however good or bad a genetic hand a pet's been dealt, stress, nutrition and exercise can all play a part. And, as with humans, we're also starting to understand that the gut and microbiome are likely to play a huge role. Deficiencies in gut bacteria, for instance, have been linked to neurological issues in dogs and small cell lymphoma in cats.

"There's some evidence that dogs in particular are born with a compromised microbiome," says Anna Webb, a canine nutrition and behaviour specialist. "And also that unlike, say, horses, which tend to bounce back pretty well after a course of antibiotics or medication, dogs' microbiomes don't. And part of this might come from ongoing diet choices - where, for instance, owners are opting for a convenience-style diet that we're increasingly understanding may not contain what dogs really need to thrive. Dogs seem to do very well eating raw meat that's unadulterated, packed with the moisture and juices that they thrive on."

It's surprisingly difficult to find good quality data on disease incidence and lifespan in dogs and cats - even in the US, electronic record systems are fragmented across different vets, and some still use paper. But what seems uncontestable is that the things that are good for us as humans - eating a balanced diet, getting plenty of exercise and outdoor time, feeling valued and loved - are also good for our shorter-lived companions, in ways we're just starting to understand.

'Particularly with senior pets, cognitive health is just as important as physical health.' Photograph: Toms Kalniņš/EPA

The microbiome, for instance, seems to be topped up by being outdoors, while studies suggest that dogs share stress levels with their owners. "There are several instances on record of dogs in Australia that lived into their mid-20s," says Webb. "These were dogs that just lived on sheep farms, running around outdoors, having a blast, eating things like cows' tails and whatever they found on their travels. A real dog's life, if you like." And in the way that humans seem to benefit from moderation, movement and ways to de-stress, it's likely the same thing happens in other mammals. "As a veterinarian with a focus on animal nutrition, I do find the development of longevity pill regimes for dogs intriguing, and I'm cautiously optimistic on the better-evidenced options," says Dr Michael Thompson, founder of Pets Food Safety, an organisation dedicated to providing dietary advice for pets. "But until longevity pills become more widely approved and available, I advocate for evidence-based approaches to improve a pet's lifespan. Ensuring pets receive a balanced diet tailored to their specific life stages, including adequate protein intake, essential for maintaining muscle mass, is crucial."

"Maintaining an active lifestyle helps keep pets at a healthy body weight and prevents obesity, which is a significant factor in various diseases - and particularly with senior pets, cognitive health is just as important as physical health," he says. "Engaging toys, training, and environmental enrichment can keep pets mentally sharp. Oral health is often overlooked, but dental disease can lead to systemic health issues. Regular tooth brushing and veterinary dental cleanings can extend a pet's life. In my practice, I've seen that when these basics of care are adhered to, pets tend to enjoy not just longer lives, but ones characterised by improved health and vitality."

But the biggest question remains: if you *can* give your pet longer life with an injection or a pill, should you? It probably goes without saying that any intervention is only worthwhile if it increases an animal's healthspan - the proportion of life they're active and healthy - in lockstep with lifespan, but even then, is a few more years at the tail end of a pets' life really for them, or for you? For me, though it's easy to say right now, I doubt I'll invest in extending Frankie's life beyond what he's likely to naturally get. As a man living in the UK, I'll be very happy if I make it to the end of my expected 78.8 years in relatively decent shape - and if my dog makes it to 15 or so with most of his joie de vivre intact, I feel like I'll have done my job.

I can also see it from other points of view. "If I had the money, I would absolutely look into interventions like these for some of our animals," says Holly Brockwell, who runs a no-kill shelter for disabled, elderly and unloved cats. "Sometimes you get one who's had a really hard life and by the time you get them they're on their last legs and you just want to give them some more time being happy and loved. I did chemotherapy on one of my ancient mogs for that reason when she got cancer, but to be quite honest, it wasn't worth the side-effects. This sounds like it might be an alternative."

Other people, at different stages of their lives, might need the companionship of their best friends in ways that I (hopefully) won't, or find the pain of replacing them too much to do it often. I certainly sympathise. But that's the tragedy and beauty of life, that everything ends. No longevity treatment is promising to let pets live as long as their owners, and there are always going to be sad goodbyes. But the important thing for all of us, humans, cats, dogs, and even guinea pigs, is to live our best lives, for as long as we can.

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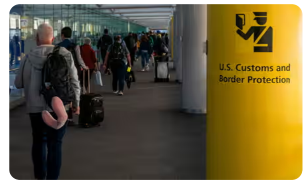

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